

Inventory of veterinary syndromic surveillance initiatives in Europe (Triple-S project): Current situation and perspectives



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ABSTRACT

Within the current context that favours the emergence of new diseases, syndromic surveillance (SyS) appears increasingly more relevant tool for the early detection of unexpected health events. The Triple-S project (Syndromic Surveillance Systems in Europe), co-financed by the European Commission, was launched in September 2010 for a three year period to promote both human and animal health SyS in European countries. Objectives of the project included performing an inventory of current and planned European animal health SyS systems and promoting knowledge transfer between SyS experts. This study presents and discusses the results of the Triple-S inventory of European veterinary SyS initiatives.

European SyS systems were identified through an active process based on a questionnaire sent to animal health experts involved in SyS in Europe. Results were analyzed through a descriptive analysis and a multiple factor analysis (MFA) in order to establish a typology of the European SyS initiatives.

Twenty seven European SyS systems were identified from twelve countries, at different levels of development, from project phase to active systems. Results of this inventory showed a real interest of European countries for SyS but also highlighted the novelty of this field. This survey highlighted the diversity of SyS systems in Europe in terms of objectives, population targeted, data providers, indicators monitored. For most SyS initiatives, statistical analysis of surveillance results was identified as a limitation in using the data.

MFA results distinguished two types of systems. The first one belonged to the private sector, focused on companion animals and had reached a higher degree of achievement. The second one was based on mandatory collected data, targeted livestock species and is still in an early project phase.

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The exchange of knowledge between human and animal health sectors was considered useful to enhance SyS. In the same way that SyS is complementary to traditional surveillance, synergies between human and animal health SyS could be an added value, most notably to enhance timeliness, sensitivity and help interpreting non-specific signals.

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1. Introduction

Traditional epidemiological surveillance systems aim to identify the presence of a particular pathogen in a population by screening the population at risk, tracking potential cases and carrying out relevant laboratory tests. These systems can be effective for the diseases they were designed for but may not be able to detect new health threats. The current context (globalization, global warming, animal and human population growth, bioterrorism threat, etc.) is favourable to emergence of new diseases (Jones et al., 2008). Health threats to come are unpredictable and even though potential threats could be identified, resources are too limited to implement systems dedicated to each of them. Facing this issue, epidemiologists have begun to explore new approaches in surveillance. One of them, called syndromic surveillance (SyS), draws more and more attention.

SyS can be defined as the real-time (or near real-time) collection, analysis, interpretation, and dissemination of health-related data to enable the early identification of the impact (or absence of impact) of potential human or animal health threats that require public and/or animal health action (Triple-S Project, 2011).

SyS was initially developed in the late 1990's by USA authorities who were looking for a way to detect bioterrorist attacks as early as possible (Lazarus et al., 2001). In Europe, interest for SyS increased after the 2003 heat wave which led thirteen Member States to implement automated mortality and emergency surveillance systems (Josseran et al., 2006; Kanieff et al., 2010). Since then, whereas several programmes (mainly in human but also animal health) have been implemented or are under development, little information on them is available. The Triple-S project¹ (Syndromic Surveillance Systems in Europe), co-financed by the European Commission and involving twenty four organizations from fourteen countries, was launched in September 2010 with the following objectives:

- Performing an inventory of existing and planned SyS systems in Europe both in animal and human sectors.
- Building a network of experts involved in SyS.
- Producing guidelines to implement SyS systems in member states.
- Proposing a European strategy to enhance collaboration between European SyS systems.
- Developing synergies between human and animal health SyS systems.

Inventories of human health SyS initiatives have already been conducted in the United States in 2008 (Buehler et al., 2008) and, for systems based on mortality, in Europe in 2011 (Kanieff et al., 2010). A literature review of the current initiatives in animal health was performed in 2011 (Dorea et al., 2011). Because the SyS approach is relatively new, most of the initiatives are not described in scientific papers. That is why the Triple-S project aimed at conducting an inventory based on a questionnaire sent to animal health specialists in Europe. This study describes the main features of the systems and initiatives identified, and the potential improvements that could be made to SyS. The potential synergies with human health and the more general overview of European veterinary SyS are discussed.

2. Materials and methods

2.1. Survey

The inventory was based on a survey conducted in two steps. The first step consisted in disseminating to people potentially involved in SyS a brief questionnaire (appendix 1) together with a brief presentation of the Triple-S project and the Triple-S SyS definition as presented in the introduction section (Triple-S Project, 2011).

The addressees were identified through a grey and white literature review pertaining to veterinary SyS. The literature was searched for on public scientific databases (Pubmed and Scencedirect) as well as on Google using advanced, customized search engine, and food safety agencies websites (appendix 2). This list was completed with official contacts in animal health, including EFSA (European Food Safety Authority) focal points, Chief Veterinary Officers (CVO), members of the European college of veterinary public health, members of the EFSA Animal Health and Welfare scientific panel and informal contacts of the Triple-S project partners.

Only brief questionnaire answers that were consistent with the Triple-S SyS definition were kept. Four types of SyS systems were considered according to their level of development: “active systems” which were already implemented and in use; “pilot systems” which were in a scoping or development phase; “completed systems” which used to be in active or pilot phase but were now completed; “explorative projects” represented the least advanced initiatives, and often corresponded to situation where syndromic data were already collected but not yet analyzed.

The second step consisted in sending the selected respondents a detailed questionnaire (appendix 3) to collect information on the main features of the system or project they were associated with. The questionnaire dealt with the general characteristics of the system: who are the data providers; how are the data collected and

¹ www.syndromicsurveillance.eu.

analyzed; how are the results disseminated; and, what are the processes for the use and evaluation of the system.

2.2. Multiple factorial analysis

The questionnaire included a lot of questions for which one or more categories could be selected. In addition to a description of each question, we wanted to evaluate if a typology of European SyS systems could be identified according to three main features: the objectives of the system, the type of data providers and the population targeted. Principal component methods are suitable to achieve this objective. Their principle is to reduce the dimensionality of multivariate datasets replacing the n original variables (active or observed variables) by p uncorrelated derived variables (principal components or factors) obtained by linear combination of active variables. Each factorial axis is orthogonal to each other (i.e., defined so that it captures the variance not explained by the previous factorial axis) and is defined by its eigenvalue which indicates the inertia (i.e., variance) of the dataset it represents (Bécue-Bertaut and Pagès, 2008; Escofier and Pagès, 2008). Active variables participate to the construction of factorial axes whereas supplementary variables are used only to aid in interpreting the results. The supplementary variables are projected onto the subspace generated by the factorial axes. We used Multiple Factorial Analysis (MFA) that allows analysis of subsets of variables (numerical, categorical or frequency variables).

The particularity of MFA, when applied to subsets of categorical variables, is to compute a global distance between units corresponding to a weighted sum of the separate distances induced by Multiple Component Analysis (MCA) performed on each subset of variables so that it balances their respective influence in this computation (Bécue-Bertaut and Pagès, 2008).

In this study we considered each question as a subset of binary variables, each binary variable corresponding to one of the question categories. Active variable categories with low figures were grouped to avoid instability in the analysis. Each variable category was represented in the factorial space defined by factorial axes.

The “objectives of the system”, the “type of data providers” and the “population targeted” were used as three subsets of binary active variables. “Motivation for transmission of data”, “Status of the system” and “Source of funding” were used as supplementary variables. The analysis was performed using FactoMineR package (Lê et al., 2008) from R software (R Development Core Team, 2010).

3. Results

3.1. Implementation of the survey

The literature review identified 14 scientific papers and ten veterinary SyS systems: Moss-Emergences in Belgium (Doherr and Audigé, 2001; Vourc'h et al., 2006); MBL cattle mortality in Italy (Vitali et al., 2009); FarmFile (Watson and Cook, 2007; Gibbens et al., 2008), SAVSNET (Radford et al., 2010) and Equine Surveillance Reports (DEFRA/AHT/BEVA, 2011) in UK; SyS on horses (Rockx et al., 2006) and

Veekijker (Bartels et al., 2006) in The Netherlands; RESPE in France (Leblond et al., 2007, 2010); VETSTAT in Denmark (Steger et al., 2003; Heuer et al., 2005); PetEpiNetVet in Romania (Radulescu et al., 2008). Using official contacts in animal health, 234 persons were identified: EFSA focal points ($n=36$); Chief Veterinary Officers ($n=27$); members of the European college of veterinary public health ($n=140$); members of the EFSA Animal Health and Welfare scientific panel ($n=21$). Ten other informal contacts were added.

A total of 248 brief questionnaires and associated letters were sent in April 2011. There were 22 responses from 13 countries: Austria(1), Belgium(3), Switzerland(1), Cyprus(1), Denmark(1), Spain(1), Finland(3), France(2), Greece(1), Italy(1), The Netherlands(3), Sweden(1) and The United Kingdom(3). From these responses, 26 contacts from ten countries were selected (some responses involved more than one contact) and sent the detailed questionnaire in July 2011. There were 18 respondents to the detailed questionnaire; allowing the identification of 25 SyS systems (some responses involved more than one SyS system). In August 2012, two other SyS systems were identified during a scientific congress. Two systems identified through literature review were not included in this study due to lack of response to the detailed questionnaire. In total, 27 different systems from 12 different countries were included (Table 1). Amongst these, eight systems were identified through literature review.

3.2. Descriptive analysis

The descriptive analysis highlighted the complexity and diversity of the European veterinary SyS systems. Indeed, more than half of the systems ($n=15$) had more than one objective, nine targeted more than one animal population, 21 had more than one data provider and 18 were monitoring more than one indicator (Tables 2 and 3).

The geographical coverage was national for 23 systems of which, ten systems covered 100% of data providers and nine systems covered 100% of the animal population under surveillance.

All systems had at least one type of data collected on an ongoing basis. Twenty-one systems used data already collected (totally or partially). Data collection for the purpose of SyS led to reorganization of the work for 13 systems, but caused no additional burden for data providers for seven systems.

Individual animal data were collected for 23 systems. About 50% of these systems collected the date of observation or registration, owner residence, animal ID Number, age, breed and sex.

For the 23 systems using clinical observations (e.g. clinical signs, lesions), 11 did not use a coding system, i.e., a closed list of clinical observations. Among the 12 systems that had a coding system, nine had their own coding system, one used a national coding system and two did not report what coding system was used.

The process to validate or not validate a statistical alarm into an epidemiological alert was similar for all systems identified. The alarm was transmitted to a relevant person (administrator or veterinarian) to validate, or not,

Table 1

European veterinary syndromic surveillance systems or initiatives identified through the Triple-S inventory conducted in 2011 and main features.

System Name (country)	Status	CA	LA	W	H	Main Objectives	Main Data providers
GMON (AU)	ACTIVE					General health surveillance,	Veterinary clinics
VETSTAT (DA)						Control of use of antimicrobials	Veterinary clinics, Pharmacies
Sikava (FI)						Outbreak detection, General health surveillance	Veterinary services, Laboratories, Farms, Slaughterhouses,
Naseva (FI)						Outbreak detection, General health surveillance	Veterinary Services, Laboratories, Farms, Slaughterhouses
REPAMO (FR)						Outbreak detection	Laboratories
SAGIR (FR)						Outbreak detection, General health surveillance	Laboratories
GD Monitor (NL)						Outbreak detection, General health surveillance	Veterinary services and clinics, Telephone help lines
FarmFile (UK)						Outbreak detection	Veterinary services
Equ. Surv. (UK)						General health surveillance	Veterinary services, Clinics and school, Laboratories
VetCompass (UK)						General health surveillance	Veterinary clinics
Kodatabasen (SW)						Production management	Laboratories, Farms, Slaughterhouses
EPI (SI)						Outbreak detection, General health surveillance	Veterinary services and clinics, Pharmacies, Laboratories, Slaughterhouses, Rendering plants
Animal Health System (SZ)	PILOTE PHASE					General health surveillance	Veterinary services and clinics, Farms, Slaughterhouses, Rendering plants,
MOSS-Emergences 2 (BE)						Outbreak detection	Veterinary services and clinics, Telephone help lines
PROVIMER (SP)						Outbreak detection, General health surveillance	Rendering plants
OMAR(FR)						Outbreak detection, General health surveillance	Rendering plants
NERGAL-abattoirs (FR)						Outbreak detection, General health surveillance	Slaughterhouses
VSD telephone log (UK)						Outbreak detection, General health surveillance	Veterinary clinics, Laboratories
Poultry practice data (UK)						General health surveillance	Veterinary clinics
AM and PM data (UK)						General health surveillance	Slaughterhouses
O48M (UK)						General health surveillance	Farms
SAVSNET (UK)						Outbreak detection, General health surveillance	Veterinary clinics, Laboratories
Kuukausi-ilmoitus (FI)	EXPLORATIVE PROJECT					General health surveillance	Veterinary services and clinics
CDB (SW)						Not define yet	Farms, Slaughterhouses
Djursjukdata (SW)						General health surveillance	Veterinary clinics and school
SVALA (SW)						Management of the diagnostic process	Veterinary clinics Laboratories, Farms, Slaughterhouses, Rendering plants
MBL(IT)	C					Surveillance of other health threats, Other	Veterinary services

CA: companion animals; LA: livestock animals; W: wild animals; H: horses; C: Completed.

using statistical information. Then, if the alarm was validated, an epidemiological investigation was initiated to determine if the alarm represented a public or animal health threat or not. An alternative process was to assess the plausibility of the alarm through an expert discussion group.

The results of surveillance were sent to data providers for 18 systems and made publicly available for seven systems.

Nine respondents declared that their systems had already been evaluated but that the evaluation mostly considered data quality rather than system performance.

Table 2

Description of active and supplementary sets of binary variables used for multiple factorial analysis on the 27 European SyS systems identified through the Triple-S inventory conducted in 2011. For each question, respondents could select more than one proposition.

	Nb. of systems concerned	Percentage (%) of systems concerned
Active sets of variables		
Objectives of the system		
General health surveillance	19	70
Detection of outbreaks	12	44
Surveillance of other threats	6	22
Other objectives	14	52
Data providers		
Veterinary clinics and schools	19	70
Veterinary services	13	48
Laboratories	11	41
Slaughterhouses	10	37
Other professional organizations	8	30
Farms	7	26
Rendering plants	6	22
Other facilities	4	15
Websites	2	7
Telephone helplines	2	7
Drug producers or pharmacies	2	7
Targeted population		
Livestock animals	23	85
Companion animals or horses	8	30
Wild animals	7	26
Supplementary sets of variables		
Status of the system		
Active	12	44
Pilot phase	10	37
Explorative project	4	15
Completed	1	4
Source of funding		
Public	22	81
Professional organization	7	26
Laboratories	4	15
Farmers	2	7
Non-profit organization	1	4
Motivation for transmission of data		
Mandatory	13	48
Access to output	12	44
Financial compensation	2	7
Other motivation ^a	13	48

^a Altruism and mutual benefit.

Seven systems already shared outputs with human health systems and three others planned to do so. Examples of synergies with human health systems described by respondents included the transmission of information about alerts, transmission of reports with interpreted data and organization of joint human and animal health meetings.

3.3. Multiple factor analysis

A description of active and supplementary variables included in the MFA is presented in Table 2. Scree test (D'Agostino and Russell, 2005) suggested keeping three factorial axes for MFA interpretation. The three groups of active variables (objectives of system, targeted population and data providers) contributed 35.4%, 44.3% and 20.3% respectively to the construction of the first factorial axis. The second factorial axis was mainly constructed by "objectives of system" and "data providers" with a contribution

of 37.7% and 55.2% respectively. The three groups of active variables contributed 33.5%, 23.0% and 43.5% respectively to the construction of the third factorial axis. The representation of active variables in the two first factorial axes space of MFA showed similar information. Only the two first factorial axes space of MFA is thus presented (Fig. 1).

The two first factorial axes space made the distinction between two groups of data providers: the first one with farms, slaughterhouses, rendering plants, other professional organizations, and veterinary services; and the second one with veterinary clinics, laboratories, drug producers or pharmacies (Fig. 1). The two first factorial axes space also distinguished livestock animals from companion animals and wild animals. Looking at supplementary variables, the two first factorial axes space opposed two groups of motivations for data providers to share their data: on one hand mandatory motivation, and on the other, financial compensation, access to outputs and other motivation. It also opposed two different states of systems:

Table 3

Answers to five selected questions of the questionnaire used for the Triple-S veterinary syndromic surveillance inventory conducted in 2011. For each question, respondents could select more than one proposition. For each question, *N* is the number of respondents.

	Nb. of respondents having ticked the proposition	Percentage (%) of respondents having ticked the proposition/the number of respondents to the questions (<i>N</i>)
Channel chosen for data transmission (<i>N</i> = 26)		
E-mail	10	38
Web portal	10	38
Surface-mail	9	35
Directly to database	7	27
Telephone	4	15
Ftp site	1	4
Transmission frequency (<i>N</i> = 26)		
Real time or near	13	50
Daily	5	19
Weekly	5	19
Other ^a	4	15
Monthly	2	8
Quarterly	2	8
Indicator monitored (<i>N</i> = 27)		
Clinical signs or symptoms	14	52
Mortality	14	52
Syndromes	13	48
Clinical diagnoses	12	44
Autopsy lesions	8	30
Laboratory test submissions	7	26
Production indicators	5	19
Other indicator	5	19
Drug prescriptions	4	15
Website hits/helpline calls	2	7
Frequency of data analysis (<i>N</i> = 27)		
Other frequency ^b	8	30
Real or near real time	7	26
Quarterly	6	22
Annually	6	22
Weekly	4	15
Monthly	4	15
Bi-annual	2	7
Daily	1	4
Statistical methods used (<i>N</i> = 23)		
No statistical method	9	39
Historical Mean	4	17
Regression model	4	17
Time-series methods	4	17
Farrington method	1	4
Control chart	1	4
Miscellaneous ^c	7	30

^a Biennial transmission or transmission only when case occurs.

^b Twice a month, on demand or biennial.

^c Spatial aggregation, *z*-test.

existing database not yet used for surveillance (“explorative project”) and active systems. In summary, the two first factorial axes space showed two groups of variables (Fig. 1) that allowed us to distinguish two types of SyS systems:

Systems based on data collected from private data providers such as veterinary clinics, laboratories, drug producers or pharmacies that targeted companion or wild animals. These systems largely had the objective to detect outbreaks or conduct general health surveillance. Systems of this group were more advanced than those of the other group.

Systems based on data collected from public stakeholders such as veterinary services, or highly regulated data providers such as slaughterhouses, rendering plants and livestock professional (farmers and professional organizations). These systems targeted livestock animals with the objective of surveillance of particular health threats, or had other objectives such as classifying farms according to non-specific health indicators. Data used for these systems were collected through a mandatory process. Systems of this group were mostly in an exploratory phase with no automated statistical analyses being performed.

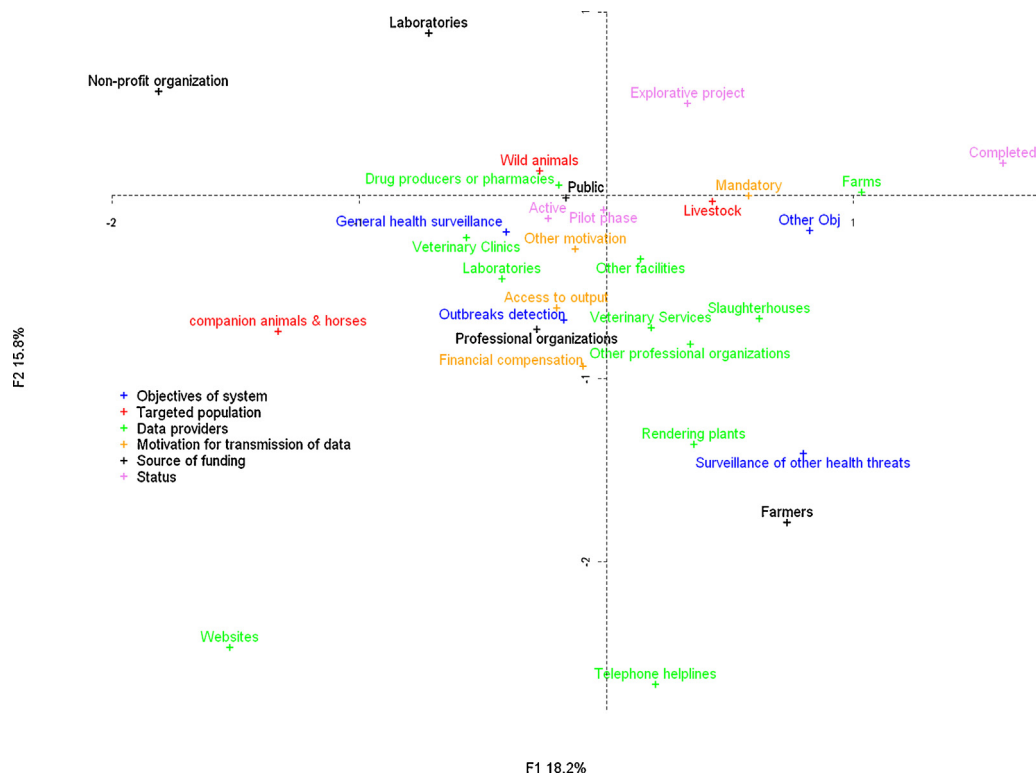


Fig. 1. Representation of modalities equal to 1 of active and supplementary variables of Multiple Factor Analysis (MFA) in the two first factorial axes performed on characteristics of European syndromic surveillance systems identified through the Triple-S inventory conducted in 2011.

4. Discussion

4.1. Method of inventory

Among the 27 SyS systems or initiatives identified, only eight were previously identified through literature review. It showed the relevance of the active approach adopted (i.e., conducting a survey). Indeed, as animal health SyS is quite recent, few scientific publications on active systems exist and an inventory based only on literature review may overlook many initiatives. Even if a lot of databases that could be used for SyS have been in existence for a long time, the idea to use them for SyS purposes is quite new. The inventory was not comprehensive (with no possibility to evaluate how many initiatives were missed), but because the questionnaire was sent to many contacts, we assume that the majority of SyS systems have been identified. This survey allowed us to draft a first snapshot of the level of development of veterinary SyS in Europe.

4.2. Limits of the SyS definitions

Most of the SyS definitions consider the real time collection and data analysis as a main feature (Buehler et al., 2004; Hoinville et al., 2009; Hoinville, 2011; Triple-S Project, 2011). In most systems, data were collected at least on a daily basis but in most cases were not analyzed in real time (Table 3).

Results of this survey highlighted the variety of veterinary SyS systems or initiatives in Europe in terms of objectives, data providers and indicators monitored (Tables 1 and 3). In this context, restricting SyS definition to the early detection seems not relevant as well as restricting any SyS system to one and unique objective.

At this stage, real time analysis should probably be considered more as a goal than a requirement for implementing veterinary SyS systems in Europe. The true distinctive feature of SyS is not timeliness (which is a goal for all systems aiming at early detection) but the nature of the indicator monitored, i.e., health related indicator. For instance weekly data could be sufficient for assessment of the impact of a health event or for the control of use of antimicrobials.

4.3. Current state of veterinary SyS in Europe

27 SyS systems or initiatives in 12 countries were identified and many of them were at a very early stage of development (Table 1). However the inventory showed that increasingly data potentially suitable for veterinary SyS are now collected in Europe and that there are an increasing number of projects to use them through SyS.

Results of MFA are paradoxical: they showed that the most advanced European SyS systems were those using data voluntarily collected by private organizations (e.g., clinics, laboratories) whereas the less achieved systems were those based on mandatory collected data. A large

amount of data suitable for SyS which are currently already collected in each Member State are not yet fully used. Thus, it seems that the potential for SyS could easily be increased in the next years.

4.4. European regulation: strength for data availability and comparability

European animal health regulation requires each Member State to collect animal health data in a similar manner to guarantee food safety and traceability. One example is the national cattle register that each Member State has to implement since 2001 to gather individual cattle identification and movement data within seven days (European Parliament, 2000). This regulation ensures the rapid availability of population and mortality data and constitutes a strong basis for building SyS systems. Two systems using these data to monitor livestock mortality are currently in pilot phase: OMAR in France (Perrin et al., 2010, 2012) and PROVIMER in Spain (Arineo, 2011).

The standardization of data should allow each Member State to build similar surveillance systems and then allow the outputs to be aggregated at the European level. A European strategy to analyze these data should be possible to define as similar was done on human mortality data through the European project EUROMOMO (Euromomo, 2008).

4.5. Accessing private data

In the animal health private sector (e.g., laboratories, veterinary clinics and professional organizations), a substantial amount of potential syndromic data are already routinely collected for sanitary, economical or zootechnical reasons, but their centralization and real-time availability is often not effective for surveillance purposes. Two main reasons were identified: (i) data providers were not willing to transmit information because it was commercially sensitive and could affect their activity; (ii) data providers were multiple, using different information technology systems that inhibits the collection and centralization of data for logistical reasons.

However, difficulties related to the willingness of data providers to share syndromic data could be overcome. In Europe, the food industry is a highly regulated activity and food processors (e.g., slaughterhouses, animal product companies), unless they are private bodies, are often bound to collect and share information to guarantee food safety and traceability. But even in absence of regulation, access to data of interest can be made possible by rewarding data providers. Rewards and incentives could consist of: (1) financial compensation e.g., SIKAVA and NASEVA Finnish systems where farmers involved in the system sell pigs and cattle for a better price (Ruoho et al., 2010); (2) benchmarking and feedback on data to provide commercial advantage e.g. SAVSNET British system (Radford et al., 2010); and, (3) access to expertise such as GD-veekijker telephone helpdesk (Bartels et al., 2006) or MOSS emergencies 2 system (Barnouin and Vourc'h, 2004; Herr et al., 2009).

Logistical issues in collecting data from the private sector could also be overcome. It is obvious that recording data should not be time-consuming for data providers. Since it is necessary to avoid duplication, i.e., notifications of the same event through two different channels, a simple and efficient way to collect data is to create an interface with the software that stakeholders already use for the management of their activities. Suitable software could also be provided directly by the manager of the SyS system e.g., SAVSNET (Radford et al., 2010) and SIKAVA use an interface with software used by veterinarians for their routine practice management. However, this was possible because only a few different practice management software packages were used by data providers of the same country. The increasing size and decreasing number of professional organizations associated with the advances in information technology means that we can expect that there will be greater compatibility and integration data from available software packages and the SyS system in the future.

4.6. Potential for enhancement and limitations of veterinary SyS in Europe

Results of this inventory showed a real interest of European countries for SyS but also showed that most of these projects were at a very early stage in animal health. Most of them lacked statistical tools to properly analyze the data. For those systems where a data analysis was carried out, only a few systems have defined protocols to interpret and investigate statistical alarms.

Most systems identified in this survey used data which were not originally collected specifically for surveillance purposes. These data are certainly cheaper to access but could be of poor quality for surveillance (bias, precision, timeliness etc.). In this survey, 41% of the systems did not use a coding system (i.e., predefined list of closed items). This lack of standardization is a hindrance to harmonization and comparability of data within one system and between systems. Initiatives to elaborate a national or regional coding systems exist in European countries: Scandinavian countries (Finland, Sweden and Norway) have work in progress to elaborate a common coding system for the diagnosis and causes of death in cattle and pigs; Austria has developed a system similar to the Scandinavian one, which is also used in some parts of Germany; a coding system was published in the Danish legislation for meat inspection of pigs and cattle, and in France for cattle meat inspection; in the UK, there is a coding system for small animals clinical data which is commonly used (VENOM, 2012) and, a coding system for diagnostic laboratory submissions from farmed livestock (Gibbens et al., 2008).

The comparability of outputs between SyS systems of different countries or within a country is an important issue for the management and communication of alerts. It may be expected that the more comparable the data are, the better the spread of disease can be identified across country border.

4.7. Synergies between traditional and syndromic surveillance systems

Synergies between traditional and SyS could be relevant to optimize costs of surveillance. The detection of the emerging Schmallenberg virus in 2011 that involved two European countries was an example of how SyS and traditional surveillance could complement each other.

The first signal revealing the presence of the virus was given by a Dutch SyS system named GD – Veekijker (GD Animal health monitor); a telephone helpline available for farmers and veterinarians to get advices when facing difficult clinical cases in livestock. In this case an increased number of calls about undiagnosed illness in cattle were reported. Around the same time, several cases in dairy cattle herds were reported in Germany through traditional surveillance system. With the knowledge of undiagnosed disease reported in The Netherlands further investigations were conducted with metagenomic analyses allowing the identification of the new Schmallenberg virus (Calavas et al., 2012; Hoffmann et al., 2012).

The Dutch SyS system raised awareness of the presence of a potential new threat, but failed to identify the cause, which was identified by a more traditional surveillance system in Germany.

4.8. Synergies between human and animal health SyS systems

Human and animal health epidemiologists are facing common statistical and epidemiological issues when dealing with SyS (e.g., use of data collected for purposes other than surveillance; standardization of clinical observations; syndrome definition; anomaly detection; interpretation of unspecific signals; response to alerts) (Dupuy et al., 2013). Therefore there is a common interest for both sides in sharing their experience and knowledge to improve their systems. The results of this survey showed that 40% of identified systems already shared or have planned to share information with the human health sector. For these systems the collaboration between human and animal health sectors was based on regular meetings to discuss the outputs of the systems e.g., GD – Veekijker (GD Animal health monitor) system in The Netherlands (Bartels et al., 2006), Farmfile system in the UK (Gibbens et al., 2008). Collaborations with human health sectors were mainly focused on zoonotic diseases.

In particular, there could be benefit in sharing knowledge to improve SyS systems performances on both sides in terms of timeliness, sensitivity and awareness. Timeliness and sensitivity for detecting a threat common to human and animal health can be better on one or the other side, depending on which species develop symptoms stronger and earlier after exposure (e.g., animal sentinel). West Nile disease surveillance is an interesting example for such synergies in European countries. In The Netherlands and France, the West Nile SyS is based on the notification by veterinarians of neurological syndromes associated with previous fever (Rockx et al., 2006; Leblond et al., 2007). In Europe, it has been observed that cases of West Nile virus in horses usually precede human cases (Leblond et al.,

2007, 2010) but the opposite has been observed in the USA and Canada (Corrigan et al., 2006). The increase of crow mortality monitored in the USA (Eidson et al., 2001, 2005; Witt, 2003) associated with West Nile virus outbreaks was not observed in Europe (Durand et al., 2002, 2005; Guptill et al., 2003). These observations highlight the value of synergy between human and animal health SyS systems for the early detection of zoonotic diseases but also the necessity to adapt SyS systems according to epidemiologic characteristics depending on geographical areas.

One of the major issues of SyS is the production of non-specific alarms that need to be investigated. Simultaneous alerts from both human and animal systems may add confidence in a signal suggesting the presence of a health threat and improve the specificity of the surveillance.

5. Conclusion

SyS is more advanced in human health than in animal health perhaps due to the greater resource available in human health. Even though none of the systems considered as SyS in this survey fulfilled all the requirements of the Triple-S SyS definition, the veterinary SyS field is promising. Indeed the diversity of data available for animal health SyS and the interest of veterinary epidemiologists for SyS highlighted by this study gives confidence in the future development of this field.

The statistical analysis was one of the weak points identified in most of the veterinary SyS systems initiatives. However, the Triple-S guidelines for the implementation of SyS systems, especially the data analysis section, and the network of people involved in SyS identified through the Triple-S project could help to enhance and spread the development of veterinary SyS in European countries. These guidelines are currently in preparation and will be available in October 2013 on the Triple-S website (<http://www.syndromicsurveillance.eu/>).

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.prevetmed.2013.06.005>.

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